Assignment 8

Josh Suber
RIN: 662038070

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1 Exercise 4.3

(a) Assume H is fixed and we increase the complexity of f. Will deterministic noise in general go up or down? Is there a higher or lower tendency to overfit?

Answer: If H (the hypothesis set or model complexity) is fixed and we increase the complexity of f (the target function), the deterministic noise will generally **increase**. Deterministic noise refers to the difference between the true target function and the best approximation achievable by the chosen hypothesis set H. When the target function f becomes more complex than the hypothesis set can capture, there will be a higher mismatch between the target function and the hypothesis set, thus increasing deterministic noise.

Tendency to overfit: With higher deterministic noise due to the model's inability to fully capture f, the model may overfit to training data in an attempt to reduce error, thus making it more likely to overfit.

(b) Assume f is fixed and we decrease the complexity of H. Will deterministic noise in general go up or down? Is there a higher or lower tendency to overfit?

Answer: If f is fixed and we **decrease** the complexity of H, the deterministic noise will generally **increase** again, since the simpler model H has less capacity to approximate the fixed target function f. However, there's a balance here:

- **Increased deterministic noise** suggests a tendency to fit the data poorly overall.
- **Reduced model complexity** makes the model less flexible and less capable of fitting random noise in the data.

These two factors influence overfitting in opposite directions. However, because the model is simpler, it is less likely to overfit by capturing noise. In this scenario, the **tendency to overfit decreases** because a simpler model is less flexible and hence more robust against fitting noise in the training data.

2 Exercise 4.5

A more general soft constraint is the Tikhonov regularization constraint

$$w^T \Gamma^T \Gamma w \leq C$$

which can capture relationships among the w_i (the matrix Γ is the Tikhonov regularizer).

(a) **What should Γ be to obtain the constraint $\sum_{q=0}^{Q} w_q^2 \leq C$?**

Answer: To obtain the constraint $\sum_{q=0}^{Q} w_q^2 \leq C$, we can set Γ to be the identity matrix I_{Q+1} of dimension $Q+1$. This gives:

$$w^T \Gamma^T \Gamma w = w^T I_{Q+1}^T I_{Q+1} w = w^T w = \sum_{q=0}^{Q} w_q^2 \leq C$$

(b) **What should Γ be to obtain the constraint $\left(\sum_{q=0}^{Q} w_q\right)^2 \leq C$?**

Answer: To obtain the constraint $\left(\sum_{q=0}^{Q} w_q\right)^2 \leq C$, we can set Γ as a $(Q+1) \times (Q+1)$ matrix where the first row contains all 1's and all other elements are 0. Thus:

$$w^T \Gamma^T \Gamma w = \left(\sum_{q=0}^{Q} w_q\right)^2 \leq C$$

3 Exercise 4.6

We have seen both the hard-order constraint and the soft-order constraint. Which do you expect to be more useful for binary classification using the perceptron model? *[Hint: sign($w^T x$) = sign($\alpha w^T x$) for any $\alpha > 0$].*

Answer: The **hard-order constraint** is expected to be more useful for binary classification using the perceptron model. In binary classification, we are only concerned with the sign of $w^T x$ to determine class labels. A soft-order constraint, which limits the magnitude of w, does not restrict the model's ability to separate classes since scaling w by a positive scalar α does not change the decision boundary (i.e., $\text{sign}(w^T x) = \text{sign}(\alpha w^T x)$ for any $\alpha > 0$).

On the other hand, a hard-order constraint can restrict certain weights by enforcing exact values, potentially limiting the model's ability to overfit by reducing the hypothesis space. This makes it more effective in controlling model complexity for binary classification, thereby reducing the risk of overfitting.

4 Exercise 4.7

Fix $\tilde{g}$ (learned from D_{train}) and define $\sigma^2_{\text{val}} \stackrel{\text{def}}{=} \text{Var}_{D_{\text{val}}}\left[E_{\text{val}}(\tilde{g})\right]$. We consider how σ^2_{val} depends on K. Let

$$\sigma^2(\tilde{g}) = \text{Var}_x[e(\tilde{g}(x), y)]$$

be the pointwise variance in the out-of-sample error of $\tilde{g}$.

(a) **Show that $\sigma^2_{\text{val}} = \frac{1}{K}\sigma^2(\tilde{g})$.**

Since (x_n, y_n) is independent from data (x_m, y_m), we have:

$$\sigma^2_{\text{val}} = \text{Var}_{D_{\text{val}}}\left[E_{\text{val}}(\tilde{g})\right]$$

$$= \text{Var}_{D_{\text{val}}}\left[\frac{1}{K}\sum_{x_n \in D_{\text{val}}} e(\tilde{g}(x_n), y_n)\right]$$

Using the properties of variance, this becomes:

$$= \frac{1}{K^2}\left[\sum_{x_n \in D_{\text{val}}} \text{Var}(e(\tilde{g}(x_n), y_n)) + \sum_{x_n, x_m \in D_{\text{val}}, n \neq m} \text{Cov}(e(\tilde{g}(x_n), y_n), e(\tilde{g}(x_m), y_m))\right]$$

Since the errors are independent, the covariance terms are zero, leaving:

$$= \frac{1}{K^2}\sum_{x_n \in D_{\text{val}}} \text{Var}(e(\tilde{g}(x_n), y_n))$$

$$= \frac{1}{K^2} \cdot K \cdot \text{Var}\left[e(\tilde{g}(x), y)\right]$$

$$= \frac{1}{K}\sigma^2(\tilde{g})$$

(b) **In a classification problem, where $e(\tilde{g}(x), y) = [\tilde{g}(x) \neq y]$, express $\sigma^2_{\textbf{val}}$ in terms of $P[\tilde{g}(x) \neq y]$.**

In a classification setting, we define $e(\tilde{g}(x), y) = [\tilde{g}(x) \neq y]$, and let $p = P[\tilde{g}(x) \neq y]$.

Since $e(\tilde{g}(x), y) = [\tilde{g}(x) \neq y]$,

$$P(e(\tilde{g}(x), y) = 1) = p$$

$$P(e(\tilde{g}(x), y) = 0) = 1 - p$$

$$E(e(\tilde{g}(x), y)) = 1 \cdot p + 0 \cdot (1 - p) = p$$

Then, the variance of $E_{\text{val}}(\tilde{g})$ is

$$\text{Var}_{D_{\text{val}}}\left[E_{\text{val}}(\tilde{g})\right] = \text{Var}_{D_{\text{val}}}\left[\frac{1}{K}\sum_{x_n \in D_{\text{val}}} e(\tilde{g}(x_n), y_n)\right]$$

$$= \frac{1}{K^2} \cdot \text{Var}\left[e(\tilde{g}(x), y)\right]$$

$$= \frac{1}{K}\left[p(1-p)^2 + (1-p)(0-p)^2\right]$$

$$= \frac{p - p^2}{K}$$

Thus,

$$P\left(\tilde{g}(x) \neq y\right) = \frac{1}{K} P\left(\tilde{g}(x) \neq y\right)^2$$

(c) **Show that for any $\tilde{g}$ in a classification problem, $\sigma^2_{\textbf{val}} \leq \frac{1}{4K}$.**

We can modify the result from part (b) as follows:

$$\sigma^2_{\text{val}} = \frac{p - p^2}{K}$$

$$= \frac{1}{K}\left[-\left(p - \frac{1}{2}\right)^2 + \frac{1}{4}\right]$$

The maximum of σ^2_{val} occurs when $p = \frac{1}{2}$, yielding:

$$\sigma^2_{\text{val}} \leq \frac{1}{4K}$$

(d) **Is there a uniform upper bound for** $\mathrm{Var}\left[E_{\mathbf{val}}(\tilde{g})\right]$ **similar to (c) in the case of regression with squared error** $e(\tilde{g}(x), y) = (\tilde{g}(x) - y)^2$**?**

$e(\tilde{g}(x), y) = (\tilde{g}(x) - y)^2$

$$\mathrm{Var}_{D_{\mathrm{val}}}\left[E_{\mathrm{val}}(\tilde{g})\right] = \mathrm{Var}_{D_{\mathrm{val}}}\left[\frac{1}{K}\sum_{x_n \in D_{\mathrm{val}}} e(\tilde{g}(x_n), y_n)\right]$$
$$= \frac{1}{K^2} \cdot \mathrm{Var}\left[e(\tilde{g}(x), y)\right]$$
$$= \frac{1}{K}\left[p(1-p)^2(\tilde{g}(x) - y)^4 + (1-p)(0 - p(\tilde{g}(x) - y)^2)^2\right]$$
$$= \frac{p - p^2}{K}(\tilde{g}(x) - y)^4$$

Since the square error is unbounded, thus there is no uniform upper bound for $\mathrm{Var}\left[E_{\mathrm{val}}(\tilde{g})\right]$.

Answer: No, there is no uniform upper bound in this case. The squared error $e(\tilde{g}(x), y) = (\tilde{g}(x) - y)^2$ is unbounded, which means that $\mathrm{Var}\left[E_{\mathrm{val}}(\tilde{g})\right]$ can also be unbounded.

(e) **For regression with squared error, if we train using fewer points (smaller** $N - K$**) to get** $\tilde{g}$**, do you expect** $\sigma^2(\tilde{g})$ **to be higher or lower?**

Answer: We expect $\sigma^2(\tilde{g})$ to be higher. With fewer training points, $\tilde{g}$ is more likely to overfit to the limited data, resulting in a higher expected error and, consequently, a higher variance. For continuous, non-negative random variables, higher mean error often implies higher variance.

(f) **Conclude that increasing the size of the validation set can result in a better or a worse estimate of** $E_{\mathbf{out}}$**.**

Answer: Increasing the size of the validation set (increasing K) decreases $\sigma^2_{\mathrm{val}} = \frac{1}{K}\sigma^2(\tilde{g})$, which reduces the variance of the estimate of E_{out}. However, this also reduces the training set size, which could lead to a less accurate model $\tilde{g}$ with higher bias. Thus, increasing the validation set size can result in either a better or worse estimate of E_{out}, depending on the trade-off between reduced variance in estimation and potential increase in model bias.

5 Exercise 4.8

Is E_m an unbiased estimate for the out-of-sample error $E_{\text{out}}(g_m)$?

Answer: Yes, E_m is an unbiased estimate for the out-of-sample error $E_{\text{out}}(g_m)$. The reason is that E_m is computed from a validation set that is independent of the training set used to create g_m. Since the validation set provides a representative sample of the data distribution, E_m serves as an unbiased estimate of $E_{\text{out}}(g_m)$, as it does not incorporate any data that was used in training, avoiding any estimation bias.